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Computer Structure

Pre-Lab Report 4

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1 Adjustable Blink

Following code would do it. The led would blink forever by the user on-time and off-time.

```
from machine import Pin
from time import sleep_ms

led = Pin(25, Pin.OUT)

on_time = int(input("Enter LED ON time (ms): "))
off_time = int(input("Enter LED OFF time (ms): "))

while True:
    led.on()
    sleep_ms(on_time)

    led.off()
    sleep_ms(off_time)
```

2 Adjustable Bouncer

It turns the LED on and off in the given period.

```
from machine import Pin
import time
from time import sleep_ms

led = Pin(1, Pin.OUT)

bounce_time = int(input("Enter Bouncing time (us): "))

my_list = []
for i in range(100):
    my_list.append((i % 7) % 3)
t1 = time.ticks_us()
for x in my_list:
```

```
t2 = time.ticks_us()
if (time.ticks_diff(t2,t1)) > bounce_time:
    led.on()
    break
if (x % 2) == 0:
    led.on()
    time.sleep_us(bounce_time//20)
else:
    led.off()
    time.sleep_us(bounce_time//20)
```

3 Reading GPIO

```
from machine import Pin
import time

led = Pin(25, Pin.OUT)

button = Pin(2, Pin.IN, Pin.PULL_UP)

while True:
    if button.value() == 0:
        led.on()
    else:
        led.off()
    time.sleep_ms(10)
```

4 Debounce

```
from machine import Pin
import time

button = Pin(2, Pin.IN, Pin.PULL_UP)

state_changes = 0
```

```
last_state = button.value()

print("Ready. Press the button to start counting.")

while True:
    time.sleep_ms(5)
    current_state = button.value()

    if current_state != last_state:
        time.sleep_ms(20)

        if button.value() == current_state:
            state_changes += 1
        print(f"State changes counted: {state_changes}")
    last_state = current_state
```